
WBI13 — Rapid Revision

IAL Biology Unit 3: Practical Skills in Biology I

Last-minute cram sheets — designed to be scanned, not read

Skill priority ordered by frequency across 21 past papers (May 2019 – May 2026)

50 marks · 80 minutes · 3 questions

All three questions equally weighted (~16 marks each)

TRIAGE — what to review, in what order

Time left	Cover these	Why
30 min	Sheets 1, 2, 6, 10 — graph criteria, graph type, calculations, phrase triggers	Graphs and calculations are in 21/21 papers. The graph is 4 marks against named criteria.
2 hours	Add sheets 3, 4, 5 — variables and design, describing a method, techniques	"Describe a method" in 20/21; "explain why" design questions ~2.9 per paper.
Half a day	Add sheets 7, 8, 9 — statistics, evaluation, risk	Fewer papers each, but each is a small self-contained block.

THREE THINGS SPECIFIC TO THIS PAPER **1. No mark ramp.** Q1 ≈ 15.8, Q2 ≈ 16.7, Q3 ≈ 17.5. No easy start, no reason to rush.
2. You have time. 1.6 min/mark vs 1.1 in Units 1 and 2. Spend six minutes on the graph.
3. No statistical tests to compute. t-test in 0/21 papers, correlation coefficient 0/21. But "significant" appeared 37× in the mark schemes — you **interpret** spread, you don't calculate tests.

SHEET 1 — THE GRAPH MARKS (21/21 papers · usually 4 marks)

Code	Criterion	What earns it
A	Axes correct	Independent variable on x, dependent on y. Where required, start at zero with no break .
L	Labels	Both axes labelled, with units. Copy label and unit exactly from the results table.
P	Plotting	All points plotted accurately on suitable linear scales . Avoid 3s and 7s; use at least half the grid.
S	Scale / line / chart type	Suitable line of best fit , or correct bar chart format, or linear scale from zero with no interruption .
K	Key	Awarded when more than one data set is plotted.

ERRORS THE MARK SCHEMES NAME **Dot-to-dot** instead of a line of best fit · axes the wrong way round · broken or interrupted axis where not allowed · **missing units** · scale using less than half the grid · no key with two data sets · plotting **raw** data when the table gives a **mean**, or the reverse.
A line of best fit is a **smooth curve or straight line** with roughly equal numbers of points either side.

SHEET 2 — GRAPH TYPE & TABLES

Independent variable is...	Draw	Key feature
Continuous — concentration, time, temperature, pH	Line graph + line of best fit	Smooth; not dot-to-dot
Categoric — species, mineral missing, extract type	Bar chart	Gaps between equal-width bars; y from zero
Continuous, grouped into classes	Histogram	No gaps — bars touch
Two measured variables, testing a relationship	Scatter graph	For correlation, not effect

DRAWING A RESULTS TABLE

- **Independent variable in the first (left) column;** dependent variable to its right
- **Headings with units** — units in the **heading only**, never beside each value
- A column per **repeat**, plus a **mean** column, plus the processed value (% change, rate)
- All values in a column to the **same number of decimal places**, matching instrument resolution
- Ruled lines, full borders, no data in the headings

SHEET 3 — VARIABLES & DESIGN (14/21 papers · ~2.9 "explain why" per paper)

Term	How to spot it
Independent variable	What the investigator deliberately changes — what differs between treatments
Dependent variable	What is measured — ask what number ends up in the results table
Control variables	Kept constant so the effect of the IV can be isolated

"**EXPLAIN WHY X WAS KEPT CONSTANT**" — **TWO PARTS, BOTH NEEDED**

1. What X would otherwise affect, **with the mechanism** — "temperature affects **kinetic energy** and therefore enzyme activity".

2. Why that matters — "so any change in the DV could **not be attributed to the IV**, and the results would not be **valid**".

"To make it a fair test" alone scores nothing.

Control this	Because it affects...
Temperature	Kinetic energy → rate, enzyme activity, diffusion
pH	Ionic and H bonds in tertiary structure → active site shape
Enzyme / substrate concentration	Number of E-S complexes formed per unit time
Volume of solution	Concentration, and surface area of tissue in contact
Size / surface area / mass of tissue	Rate of diffusion in or out
Time	Longer exposure → larger change regardless of treatment
Age / variety / source of organism	Genetic and physiological differences

CONTROL EXPERIMENT ≠ CONTROL VARIABLE

A control experiment runs **without** the factor being tested. Standard answer: "discs soaked in the **solvent alone**, so any effect is due to the **extract and not the solvent**, giving a baseline." Other forms: boiled/denatured enzyme; distilled water in place of the test solution.

SHEET 4 — THE RECURRING "EXPLAIN WHY" ANSWERS

Question	Answer
Why 25 °C / 30 °C?	Close to the optimum → measurable rate; above room temperature so not too slow. In school microbiology, 25 °C not 37 °C reduces risk of culturing human pathogens .
Why maintain pH?	pH affects tertiary structure → active site shape. Use a buffer .
Why measure initial rate?	Substrate concentration is highest and not yet limiting → rate is maximal. Later it falls as substrate is used up, so a later reading depends on how long you waited.
Why a fixed incubation time?	So time is a control variable — all treatments compared over the same period.
Why let the solvent evaporate?	So the solvent itself doesn't inhibit the bacteria and affect the zone.
Why repeat?	Identify anomalies · calculate a mean → reduces random error · allows a standard deviation

FIVE TERMS, USED CORRECTLY

Accurate = close to the true value · **Precise** = repeats close to each other (can be precise but inaccurate) · **Resolution** = smallest change the instrument detects · **Reliable** = same result on repetition · **Valid** = measures what it set out to, with other variables controlled. An invalid experiment can't be rescued by repeating it.

Anomalies: **identify and exclude from the mean** — say so explicitly.

SHEET 5 — DESCRIBING A METHOD (20/21 papers · top command word, 3.6/paper)

WRITE NUMBERED STEPS WITH QUANTITIES — NEVER PROSE

1. Apparatus, with the measurement and unit **2.** Quantity or concentration, stated numerically **3.** How the IV is varied, over what range **4.** How the DV is measured, and when **5.** Variables controlled **6.** Repeats and means

"Add 10 cm³" scores; "add some solution" doesn't.

SOLUTION OF KNOWN CONCENTRATION

1. Weigh a known mass on a balance (state it) · **2.** Dissolve in **less than** the final volume of distilled water, stirring · **3.** Transfer to a **volumetric flask**, make up to the mark · **4. Filter** if solid remains — needed when the DV is measured by colour or absorbance.

SERIAL DILUTION

1 cm³ stock + 9 cm³ distilled water = 1 in 10. Mix, take 1 cm³ of that, repeat. Each step dilutes by the same factor. **Dilution factor = final volume ÷ volume of stock.**

ASEPTIC TECHNIQUE — SIX POINTS WITH REASONS

- **Flame the inoculating loop** until red hot, then cool
- Work **close to the Bunsen** — updraught carries airborne microbes away
- Lift the Petri dish lid **only slightly, at an angle** — never remove it
- **Flame the neck** of culture bottles before and after pouring
- Tape the lid but **do not seal completely** — prevents growth of **anaerobic** pathogens
- Incubate at **≤25 °C**; store dishes **inverted** so condensation doesn't drip on the agar

Zone of inhibition: measure the **diameter** in mm, two readings at right angles, take the mean. Larger zone = more effective. Area = **πr²**, and r is **half** the diameter.

SHEET 6 — THE RECURRING PRACTICAL CONTEXTS (the pool is narrower than it feels)

Context	Papers	What to know
Plant tissue in sucrose / salt potato, carrot, beetroot	12/21	Cut with a cork borer (constant diameter), cut to equal length · blot dry before each weighing · fixed time, constant temperature · calculate percentage change in mass because initial masses differ · where the line crosses zero , there is no net water movement , so the solution's water potential equals the tissue's
Enzyme investigations	6/21	Control pH with a buffer , temperature with a water bath · measure the initial rate · equilibrate solutions separately before mixing
Seed germination	6/21	Equal numbers of seeds, moist filter paper, same temperature, light and water volume · DV is percentage germination or mean radicle length after a fixed time
Eye-piece graticule	6/21	See sheet 7
Vitamin C / DCPIP	5/21	DCPIP is blue → colourless when reduced by vitamin C. Titrate sample into a fixed volume of DCPIP until blue just disappears; compare with a standard solution of known concentration . Smaller volume needed = higher vitamin C content .
Plant minerals	5/21	Culture solutions each lacking one mineral; complete solution as the control ; compare height, mass or leaf colour after a fixed period
Zone of inhibition	4/21	See sheet 5
Tensile strength of fibres	4/21	Hang increasing masses until the fibre breaks. Tensile strength = force ÷ cross-sectional area ; area = πr^2 . Control fibre length, diameter and how mass is added.
Mitotic index from root tips	3/21	Hydrolyse in acid, stain, squash under a coverslip for a single cell layer, count several fields of view. Index = cells in mitosis ÷ total cells .
Beetroot membrane permeability	2/21	DV = red pigment released, measured by colorimeter as absorbance. Rinse cut discs first to remove pigment released by cutting.

SHEET 7 — MICROSCOPY & THE GRATICULE (6/21 papers)

CALIBRATION — FIVE STEPS **1.** Place a **stage micrometer** on the stage (a scale of known dimensions). **2.** Focus and **line up the eyepiece graticule against the stage micrometer**. **3.** Count how many **eyepiece divisions** match a known length on the micrometer. **4.** Divide → the length of **one eyepiece division** at that magnification. **5.** Remove the micrometer, view the specimen, count divisions and multiply.
The calibration is valid only for that objective lens — changing magnification means recalibrating. That point is often the mark.

magnification = image size ÷ actual size · 1 mm = 1000 μm = 1 000 000 nm · convert **before** dividing.

SHEET 8 — CALCULATIONS (21/21 papers, ~2.3 each)

Type	Method	Trap
Percentage change	(final – initial) ÷ initial × 100; keep the sign	Dividing by the final value; dropping the minus
Rate	change ÷ time, with units. From a curve: tangent gradient	Omitting units
Initial rate	Gradient of the tangent at t = 0	Using the overall gradient
Mitotic index	cells in mitosis ÷ total cells	Counting interphase cells
Magnification	image ÷ actual; convert units first	Not converting mm → μm
Tensile strength	force ÷ πr^2 ; r = half the diameter	Using diameter for r
Ratio	Divide both by the smaller; give in the order asked	Reversing the order
Dilution factor	final volume ÷ volume of stock	1 cm³ into 9 cm³ makes 10 total
Area of a zone	πr^2 , where the measurement is usually the diameter	Substituting diameter for r
Percentage error	(uncertainty ÷ measured value) × 100	Forgetting to double it for two-reading measurements

RULES Show every step · units and the right s.f. or d.p. · convert **before** calculating · sanity-check the magnitude.
Significant figures: give a processed value to the **same s.f. as the least precise raw data**. Never copy out a full calculator display.
Uncertainty of one reading = **half the smallest division; double it** for a measurement needing two readings (a change in mass, a length from both ends).
Percentage error is largest for the smallest measurements → using larger volumes, masses or times reduces it. Standard improvement to suggest.

SHEET 9 — SPREAD, ERROR BARS & SIGNIFICANCE ("significant" ×37 in the mark schemes)

- **Standard deviation** = spread of data **about the mean**. **Large SD** → widely spread, mean **less reliable**. **Small SD** → clustered, mean more representative.
- **Error bars** at ± one standard deviation show that spread graphically.

THE EXACT WORDING — NOTE THE HEDGING Error bars **overlap** → the difference is **probably not significant**, and may be due to chance.
Error bars **do not overlap** → the difference is **likely to be significant**.
Always add: "a statistical test would be needed to confirm it." The bars **suggest** significance, they do not prove it. That qualification is frequently the final mark.

Which test, if asked: **Student's t-test** compares the means of two groups · **chi-squared** compares observed with expected frequencies · **correlation coefficient** measures the strength of a relationship. Significant when the probability it arose by chance is < **0.05 (5%)**, i.e. the **calculated value exceeds the critical value** at the appropriate degrees of freedom.

SHEET 10 — CONCLUSIONS, LIMITATIONS & RISK (10/21 papers)

A CONCLUSION MUST

- State the relationship **using figures from the data, with units**
- Answer the **original question**
- Stay **within the range tested** — no extrapolation, no generalising to other species
- Say **correlation, not causation** where the data shows an association

Limitation	Paired improvement
Too few repeats — anomalies invisible, mean unreliable	More repeats; calculate mean and standard deviation
Small or unrepresentative sample	Larger sample, selected randomly
A variable not controlled	Name it and say how — water bath, buffer, fixed time
Instrument of low resolution	Finer graduations, or a data logger
End point judged subjectively by eye	Colorimeter , or a colour standard
Narrow range of the IV tested	Extend the range; smaller intervals near the point of interest
Biological variation between samples	Same source, same age, same variety

Every limitation must be **paired with a specific improvement**. "Do more repeats" alone is worth little; "do five repeats at each concentration and calculate a mean, so anomalies can be identified" earns the mark.

RISK ASSESSMENT — NAME HAZARD, RISK AND PRECAUTION Scalpel / cork borer — cuts — cut downwards onto a tile, away from the hand · Microorganisms — infection — aseptic technique, ≤25 °C, don't open sealed plates · Ethanol — flammable — keep from naked flames · Acids, alkalis, DCPIP, stains — irritant — goggles and gloves · Hot water baths — scalds — use tongs.
Ethics with living organisms: minimise harm and distress · use the fewest organisms necessary · return them unharmed where possible.

SHEET 11 — PHRASE TRIGGERS

If the question is about...	...the phrase they want is
Why control a variable	"So any change in the DV is due to the IV alone and the results are valid "
Why a control experiment	"To show the effect is due to the extract, not the solvent , giving a baseline "
Why repeat	"Identify anomalies , calculate a mean , reduce random error "
Why initial rate	" Substrate concentration is highest and not yet limiting, so the rate is maximal"
Why percentage change	"The initial masses differ , so percentage change allows a fair comparison "
Overlapping error bars	"The difference is probably not significant and may be due to chance"
Non-overlapping error bars	" Likely to be significant ; a statistical test would be needed to confirm"
Large standard deviation	"Data are widely spread about the mean, so the mean is less reliable "
Graticule	"Calibrate against a stage micrometer ; valid only for that objective lens "
An association in data	"A correlation ; it does not establish causation "
Describing a trend	Figures with units plus the shape — "rises from 2.1 to 6.8 between 0 and 40 min, then plateaus"
Suggesting an improvement	A specific change plus why it helps — "use a colorimeter, removing subjective judgement of the end point"
Why 25 °C in microbiology	"Reduces the risk of culturing human pathogens "
Why not seal the Petri dish	"To prevent growth of anaerobic pathogens"

SHEET 12 — 60-SECOND SELF-TEST

Cover the right column. Hesitate on one, go back to that sheet.

Question	Answer
Five graph mark codes?	A axes · L labels with units · P plotting on linear scales · S scale/line/chart type · K key
Which variable on which axis?	Independent on x, dependent on y
Continuous IV — which graph?	Line graph with a line of best fit
Categoric IV — which graph?	Bar chart, with gaps between bars
Histogram vs bar chart?	Histogram bars touch (continuous data); bar chart has gaps
Where do units go in a table?	In the heading only, never beside each value
Define a control variable.	A variable kept constant so the effect of the IV can be isolated
Control variable vs control experiment?	Control variable is held constant; a control experiment runs without the factor being tested
Why measure initial rate?	Substrate concentration is highest and not yet limiting, so rate is maximal
Why percentage change in mass?	Initial masses differ, so percentage allows a fair comparison
Accurate vs precise?	Accurate = close to the true value; precise = repeats close to each other
What is resolution?	The smallest change an instrument can detect
What do you do with an anomaly?	Identify it and exclude it from the mean
Three steps of a serial dilution 1 in 10?	1 cm³ stock + 9 cm³ water, mix, take 1 cm³ and repeat
Why not fully seal a Petri dish?	To prevent growth of anaerobic pathogens
Why incubate at 25 °C in school?	Reduces risk of culturing human pathogens
What is a stage micrometer for?	Calibrating the eyepiece graticule
Limitation of a graticule calibration?	Valid only for that objective lens
Tensile strength formula?	Force ÷ cross-sectional area, where area = πr²
DCPIP colour change with vitamin C?	Blue to colourless
Smaller titre of juice means what?	Higher vitamin C content
Mitotic index formula?	Cells in mitosis ÷ total cells
Uncertainty of a single reading?	Half the smallest division — doubled if two readings are needed
Percentage error formula?	(uncertainty ÷ measured value) × 100
Overlapping error bars mean what?	The difference is probably not significant
Large standard deviation means what?	Data widely spread; the mean is less reliable
What must every limitation be paired with?	A specific improvement
Three parts of a risk assessment answer?	The hazard, the risk it poses, the precaution

WALKING INTO THE EXAM Spend six minutes on the graph — it's four marks against fixed criteria. Write methods as numbered steps with quantities. Every "explain why" needs the mechanism **and** the consequence for validity. Hedge every significance claim. All three questions are worth the same, so pace evenly — at 1.6 minutes per mark you are not short of time.